Mesh Relay Field Tests: Layouts, Analysis and Deployment Rule

Oct 23, 2026 · @Hansika Sayyad

Summary

A direct laptop–Pi link carries every map out to 28 m (RSSI about −59 dBm), and starts missing maps near −72 dBm. A relay should therefore be dropped when the rover's RSSI, averaged over 10 s, falls to −68 dBm, with a map-age check from the base as a backstop.

We ran three straight-line test cases on 16 and 22 October 2026: 49 runs of 60 s each, 2,908 one-second rows.

1. **Direct link (scenarios 1–8):** laptop moves away from the Pi, 0–37 m.
2. **Relay range (scenarios 31–34):** relay N1 moves away from the Pi, 22–29 m, with the laptop beside it.
3. **Relay fixed (scenarios 35–41):** N1 fixed at 27.84 m, laptop moves 0–19 m beyond it.

Three conclusions drive the design:

- **The map, not TF, is what fails.** Combined PDR stayed at 91–100 % while maps stopped for up to 34 s. The score is therefore built on map gaps.
- **The relay's own link limits the two-hop route.** The N1–Pi hop collapsed from 62 % to 21 % of maps between 26.7 m and 29 m. The relay must be dropped while its link back is still strong.
- **Two hops lose maps even at good signal.** As configured, no two-hop position met the target. The map transport needs fixing (smaller DDS messages or map compression) before any threshold can protect it.

Each scenario has only 1–5 runs, so every number here rests on 1–5 minutes of data.

System setup

The laptop stands in for the exploring rover and sends the map and TF onto the mesh. The base Pi receives them and measures everything.

Node	Role	Hardware	Wi-Fi MAC
L (laptop)	Explorer stand-in; publishes map + TF	Laptop	28:A4:4A:AC:3C:1C
Pi (base)	Receiver; logs all metrics	Raspberry Pi, board A on 16 Oct	D8:3A:DD:13:54:64
Pi (base)	Receiver; logs all metrics	Raspberry Pi, board B on 22 Oct	2C:CF:67:34:1D:9C
N1 (relay)	Repeater we want to deploy	Raspberry Pi Zero 2 W	2C:CF:67:23:DE:00
ESP32	RSSI sniffer on channel 1	ESP32	–

Network: BATMAN-adv over an 802.11 IBSS cell, 2.4 GHz channel 1. ROS 2 Jazzy with Fast DDS.

Stream	QoS	Rate	Size	Load
Map	RELIABLE + TRANSIENT_LOCAL	1 every 3 s	~106 KB measured	~35 KB/s
TF	BEST_EFFORT	~33 msg/s	56 B	~2 KB/s

Why the map matters more. The base needs the map to see what has been explored and to plan. A lost TF sample is replaced by the next one 30 ms later; a lost map is not.

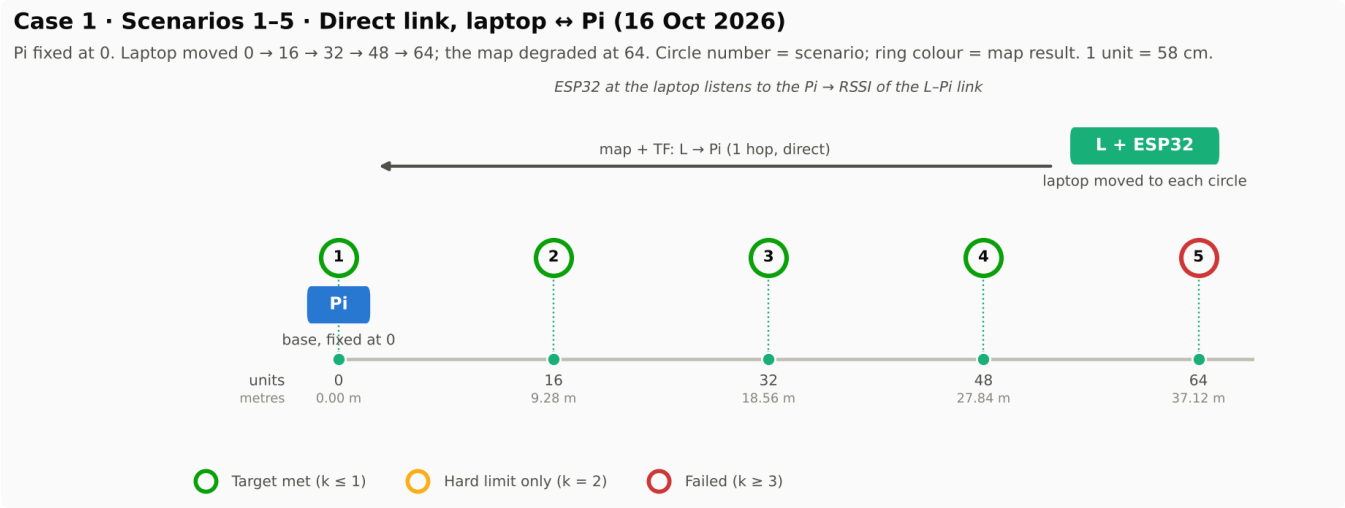
Map requirement. Missed maps in a row: k = round(gap / 3 s) − 1.

- **Target:** k ⩽ 1 (gap about 6 s).
- **Hard limit:** k ⩽ 2 (gap about 9 s). k ⩾ 3 is a failure.

Test environment and layouts

All tests ran along one straight line, with positions marked in units of 58 cm (metres = units × 0.58). In each picture the circle number is the scenario, and the ring colour is the map result from section 5.

Case 1: direct link, scenarios 1–8



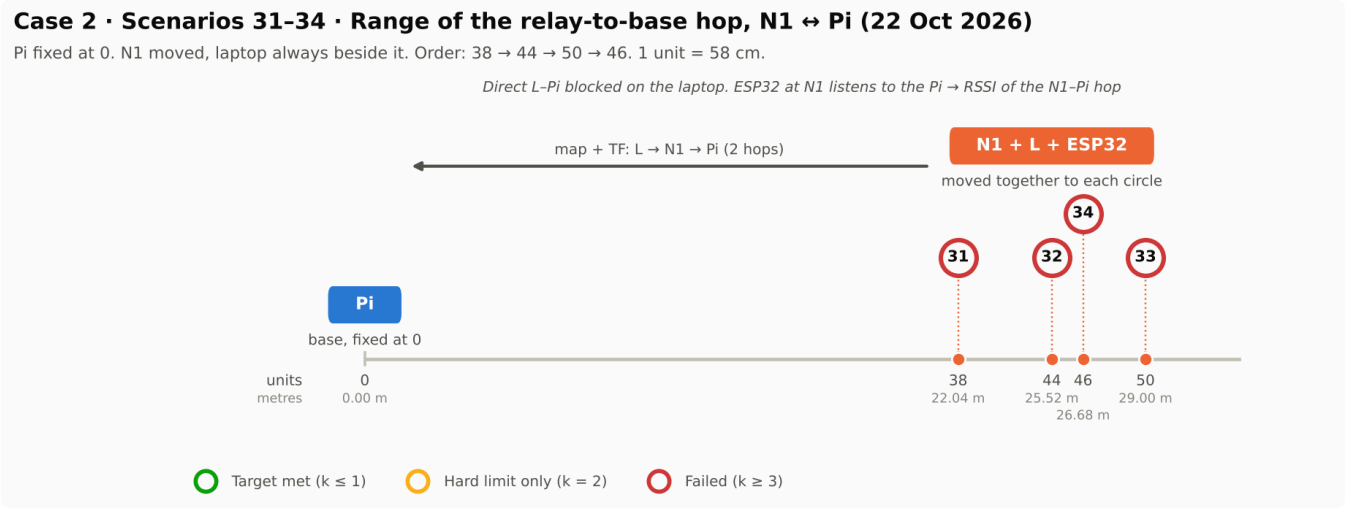
Purpose: find how far a direct laptop–Pi link carries the map.

Scenario	Laptop position (units)	Laptop–Pi distance
1	0	0 m
2	16	9.28 m
3	32	18.56 m
4	48	27.84 m
5	64	37.12 m

The notebook's "0.928 m" for scenario 2 is a slip; 16 × 0.58 = 9.28 m.

Case 1 ends at scenario 5 because the map link had already degraded there. Positions beyond 37 m (scenarios 6–8) were also recorded but are left out of this report, since a relay would be deployed before the rover reaches them.

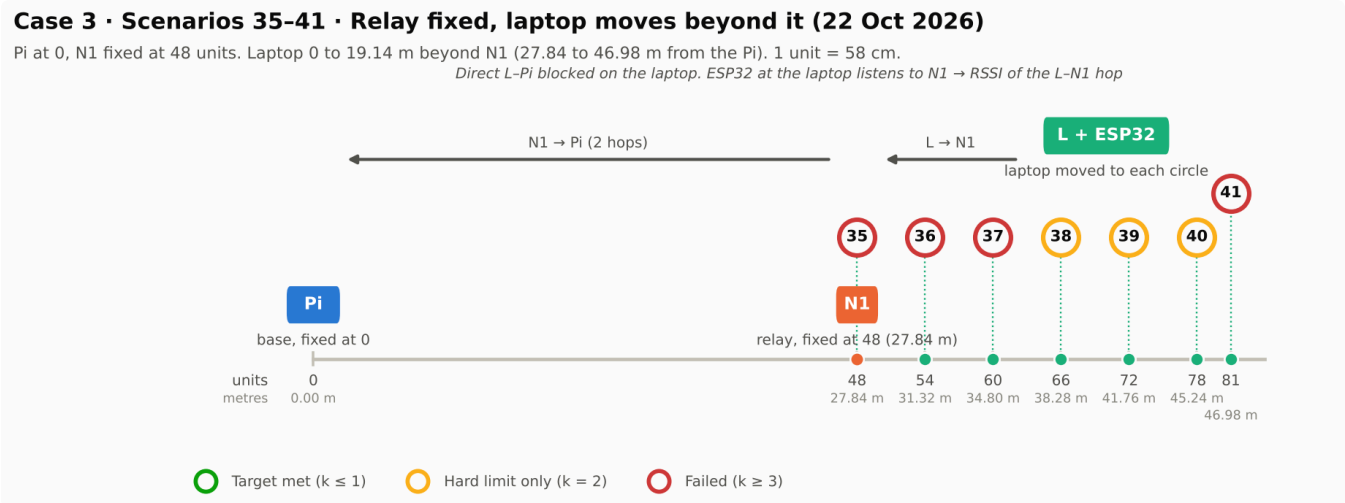
Case 2: range of the relay-to-base hop, scenarios 31–34



Purpose: find how far N1 can be from the base. The laptop and the ESP32 moved with N1, and the direct laptop–Pi link was blocked, so traffic went L → N1 → Pi.

Scenario	N1 + laptop position (units)	N1–Pi distance
31	38	22.04 m
32	44	25.52 m
33	50	29.00 m
34	46	26.68 m

Case 3: relay fixed, laptop moves beyond it, scenarios 35–41



Purpose: find how far the rover can go past a relay. N1 stayed at unit 48 (the notebook drawing shows 46; 48 is correct). The direct link was blocked, so there were 2 hops.

Scenario	Laptop position (units)	Laptop–N1	Laptop–Pi
35	48	0 m	27.84 m
36	54	3.48 m	31.32 m
37	60	6.96 m	34.80 m
38	66	10.44 m	38.28 m
39	72	13.92 m	41.76 m
40	78	17.40 m	45.24 m
41	81	19.14 m	46.98 m

Data collection

The base Pi logged one row per second for each 60 s run, giving 2,908 rows in 49 runs (scenarios 6–8 left out as above; scenario 42 excluded, as it was not a real test).

What each row holds:

- **Delivery:** TF and map received / expected, combined PDR, TF PDR, map PDR.
- **Throughput:** total, TF and map, over the window and the last second.
- **Latency:** min / avg / max, jitter, TF latency, map latency.
- **BATMAN:** hop count, TQ (link quality 0–255).
- **RSSI:** from an ESP32 sniffer on channel 1. The filtered value (median of 3 + EMA, α = 0.35) is used.
- **map_active:** 1 when a map arrived in that second. Gaps between maps come from this.

Metrics use a 6 s rolling window, so rows overlap. All results are worked out per run or per scenario, never across a run boundary.

Where the RSSI was measured (compare like with like):

Case	ESP32 at	Listening to	Link measured
1 (sc 1–5)	Laptop	Pi board A	Laptop ↔ Pi, the direct link

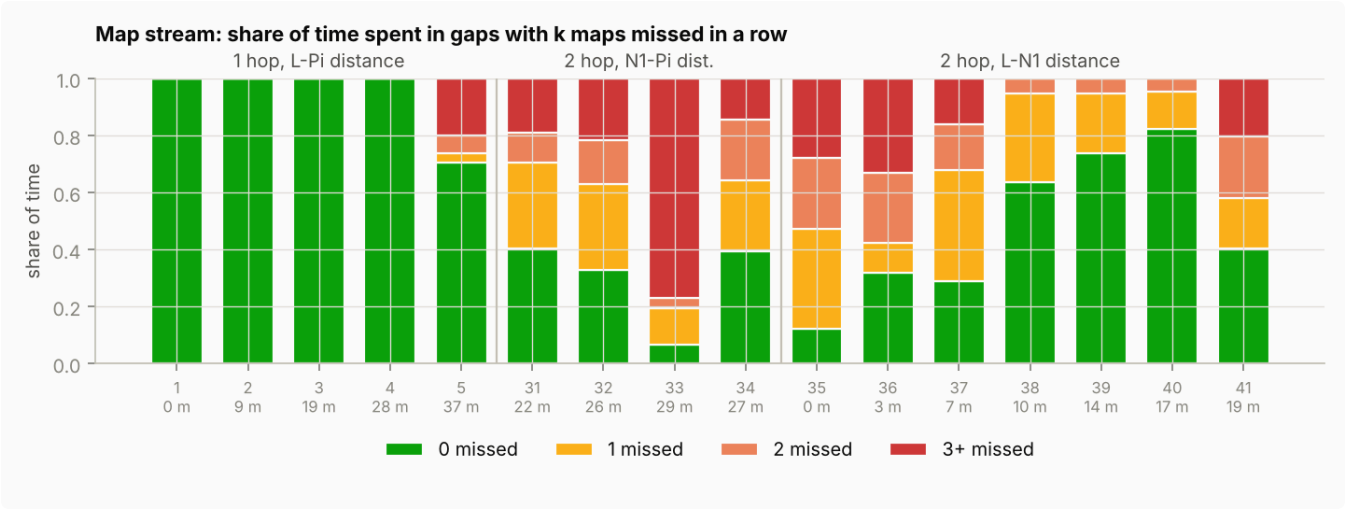
Case	ESP32 at	Listening to	Link measured
2 (sc 31–34)	N1	Pi board B	N1 ↔ Pi, the second hop
3 (sc 35–41)	Laptop	N1	Laptop ↔ N1, the first hop

Data audit:

Check	Result
Runs per scenario	1–5; most runs 60 rows. Short: 41/1 (30 rows); 5/1 and 39/1 (59 rows)
Hop count	1 in every row of case 1, 2 in every row of cases 2–3
RSSI target MAC	Matches the table above in every block
Total outages (empty PDR)	3 rows, all in scenario 41. Counted as 0 % delivery
Empty TQ	199 rows, but 196 of them still had traffic. Treated as BATMAN dropouts, not outages
Empty RSSI	9 rows (filtered value only)
Map size	~106 KB everywhere except scenario 2 (39–83 KB)
Row timing	Two-hop rows up to 1.66 s apart, so gaps were timed from timestamps

Results

Only case 1 up to 28 m met the map target. The direct link failed at 37 m (scenario 5), and every two-hop position missed at least two maps in a row.



Map stream per scenario

Delivery = maps received ÷ maps due (one per 3 s). Gap figures use whole gaps; max also counts gaps cut by the run edges.

Sc	Link length	Runs	Delivery	Gap med / p95 / max	Map latency med / p90	Result
1	0 m	1	100 %	3.0 / 3.0 / 3 s	206 / 278 ms	Target met
2	9.3 m	3	100 %	3.0 / 3.0 / 4 s	236 / 343 ms	Target met
3	18.6 m	3	100 %	3.0 / 3.0 / 3 s	365 / 441 ms	Target met
4	27.8 m	3	100 %	3.0 / 3.1 / 3 s	349 / 402 ms	Target met
5	37.1 m	3	77 %	3.0 / 5.8 / 34 s	702 / 1045 ms	Failed
31	22.0 m	3	60 %	3.0 / 10.2 / 21 s	175 / 368 ms	Failed
32	25.5 m	3	60 %	3.1 / 10.2 / 14 s	158 / 254 ms	Failed
33	29.0 m	4	21 %	5.6 / 16.3 / 57 s	207 / 386 ms	Failed
34	26.7 m	3	62 %	3.0 / 9.0 / 24 s	180 / 402 ms	Failed
35	0 m	3	45 %	6.0 / 11.7 / 14 s	262 / 716 ms	Failed
36	3.5 m	3	50 %	3.2 / 15.0 / 15 s	202 / 404 ms	Failed
37	7.0 m	3	58 %	3.2 / 9.1 / 15 s	251 / 545 ms	Failed
38	10.4 m	3	80 %	3.0 / 6.0 / 9 s	301 / 582 ms	Hard limit only
39	13.9 m	3	89 %	3.0 / 6.2 / 9 s	302 / 572 ms	Hard limit only
40	17.4 m	3	93 %	3.0 / 5.4 / 8 s	390 / 645 ms	Hard limit only
41	19.1 m	5	61 %	3.2 / 9.0 / 27 s	1244 / 2682 ms	Failed

Link length = L–Pi for 1–5, N1–Pi for 31–34, L–N1 for 35–41.

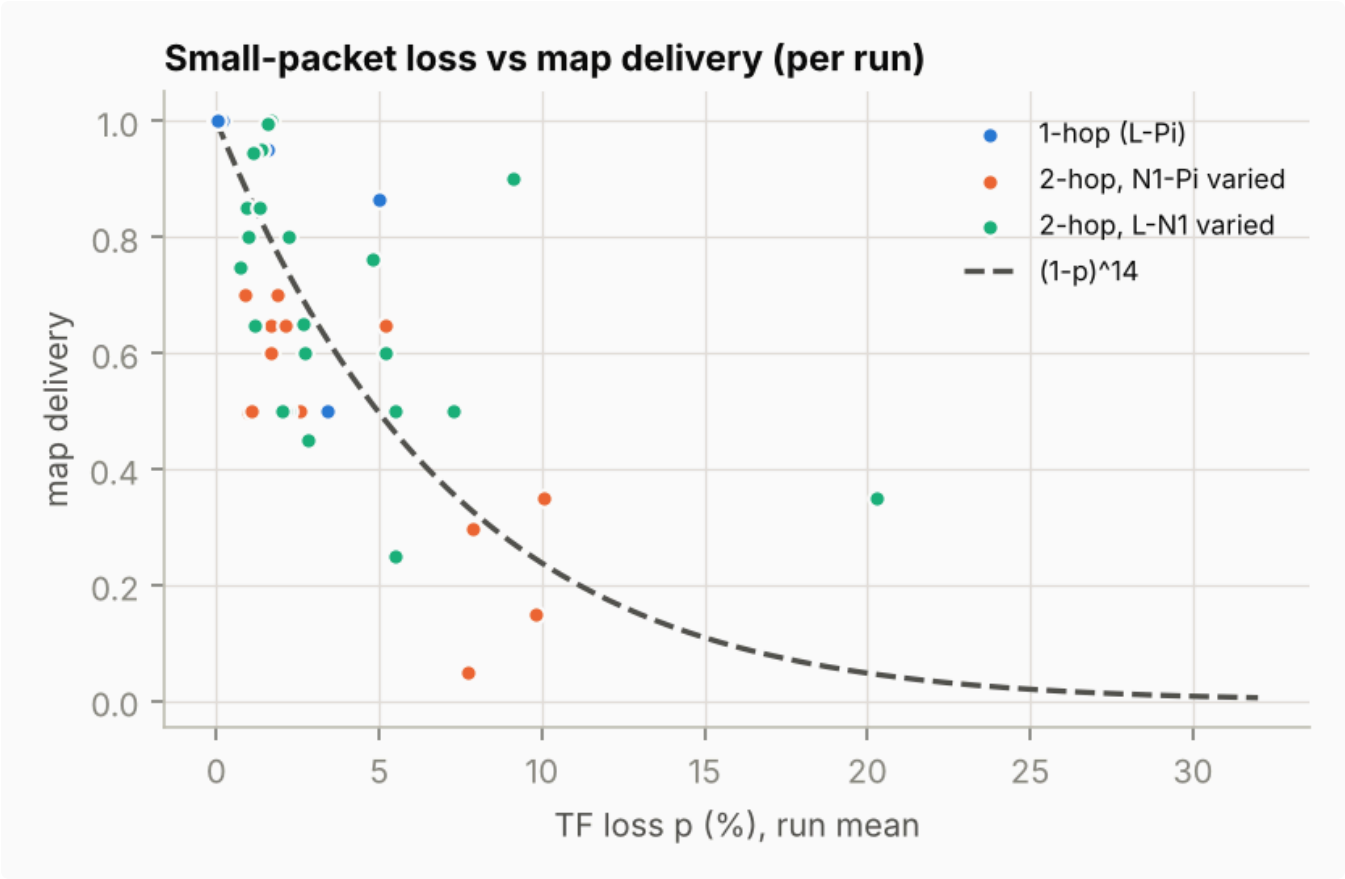
TF stream and RSSI per scenario

Sc	TF PDR mean	TF latency med	Jitter med	Combined PDR med	TQ med	RSSI med
1–4	99.9–100 %	36–67 ms	15–18 ms	100 %	237–254	−47 to −59 dBm
5	96.7 %	201 ms	24 ms	99.5 %	227	−71.7 dBm
31	98.8 %	108 ms	32 ms	99.0 %	223	−65.1 dBm
32	97.3 %	128 ms	34 ms	98.6 %	221	−67.2 dBm
33	91.1 %	162 ms	38 ms	91.4 %	207	−76.0 dBm
34	97.8 %	143 ms	35 ms	98.0 %	231	−74.2 dBm
35	94.0 %	134 ms	35 ms	95.5 %	226	−63.3 dBm
36	97.9 %	113 ms	33 ms	98.0 %	209	−74.2 dBm
37	97.8 %	114 ms	34 ms	98.0 %	221	−77.2 dBm
38	98.7 %	111 ms	32 ms	99.0 %	219	−81.8 dBm
39	97.6 %	154 ms	36 ms	99.5 %	207	−81.6 dBm
40	98.6 %	162 ms	36 ms	100 %	227	−81.2 dBm
41	92.4 %	536 ms	40 ms	98.2 %	142	−89.5 dBm

What the results show

1. Combined PDR hides map loss. Combined PDR stayed at 91–100 % while maps stopped for up to 34 s. TF makes up almost all packets, so combined PDR is really TF PDR (correlation 1.00).

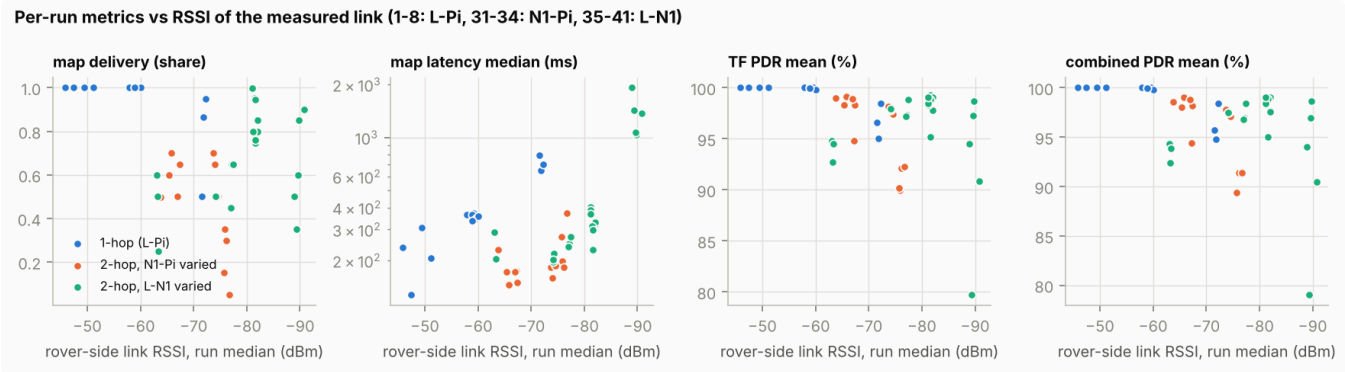
2. Small TF loss means large map loss. Across runs, map delivery follows about (1 − p)^14, where p is TF loss (correlation −0.75). About 2 % TF loss already costs about 25 % of maps. A likely reason: each 106 KB map is split into many IP fragments, and losing any one fragment loses the whole piece, which RELIABLE then resends.



3. The one-hop cliff sits between −61 and −72 dBm. At −47 to −61 dBm (0–28 m) no map was missed. At about −72 dBm (37 m), 2 of 3 runs missed two or more maps in a row, and one lost maps for 34 s. The first such gap appeared at −71.8 dBm. No position was tested between −61 and −70 dBm.

4. At the cliff, the same RSSI can pass or fail. The three runs of scenario 5 had nearly the same RSSI (−71.6 to −72.3 dBm). One kept the target (95 % of maps), one missed two in a row, and one lost maps for 34 s. Near −70 dBm the link *can* fail, so the trigger must sit above that zone.

5. Map latency warns early on one hop. It doubled from 206–365 ms (≤ 28 m) to 702 ms (37 m), at the same position where maps started being missed.



6. Two hops lose maps even at good RSSI. One hop delivered 100 % of maps at −59 dBm, but two hops delivered only 45–60 % at −63 to −67 dBm (scenarios 31, 32, 35). Near the cliff, one hop delivered 77 % at −72 dBm, against 50–62 % for two hops at −74 dBm. Jitter doubled from 15–24 ms to 32–40 ms. Part of this is built in: on one channel every map crosses the air twice. The Pi board also changed between days, which shifted RSSI by about 15 dB.

7. The relay-to-base hop is the weak link. In case 2, N1–Pi delivery fell from 62 % at 26.7 m to 21 % at 29 m. In case 3, the laptop–N1 hop still delivered 80–93 % at −81 dBm. N1 at 27.84 m was only about 1 m from its own cliff.

8. Scenarios 35–37 did worse than 38–40, even though the laptop was closer to N1. This held in every run, but delivery also rose steadily with time (13:22 → 13:52). The cause is likely outside the laptop–N1 hop: the N1–Pi hop (not logged then) or channel interference.

Parameter selection

The score keeps four parameters: map gap, map latency, TF PDR and TF latency. The rest were dropped because they repeat another parameter, don't separate pass from fail, or describe a cause rather than delivered quality.

Correlations are Spearman ρ across 49 runs.

Candidate	Decision	Why	Measured by
Map gap (missed maps in a row)	Keep	It is the requirement itself	Base
Map latency	Keep	Early warning on one hop; doubles before maps are missed	Base
TF PDR	Keep, small weight	Shows small-packet loss, which drives map loss (ρ = −0.75)	Base
TF latency	Keep, small weight	Pose freshness; reaches 536 ms in scenario 41	Base
Map throughput	Drop	Repeats map delivery (ρ = 0.80); map size varies	Base

Candidate	Decision	Why	Measured by
Combined PDR	Drop	Same as TF PDR (ρ = 1.00)	Base
Jitter	Drop	Tracks TF latency (ρ = 0.83) and hop count	Base
TQ	Drop	Lags about 60 s, often missing, failed runs had TQ 223–231	Rover
RSSI	Not in score; used as the trigger	A cause, not an outcome (see below)	Rover
Hop count	Not in score; used as context	Does not change as the link degrades (see below)	Rover

Why hop count is not in the score

- **It doesn't move while the link degrades.** It was 1 in every row of case 1 and 2 in every row of cases 2–3, at 0 m and at 37 m, in passing and failing runs.
- **It would punish the fix.** Deploying a relay raises hop count. In the score, that would lower Q, and a lower Q could ask for another relay.
- **Its effect is already counted.** Two hops show up as lower map delivery and doubled jitter, which the map terms already see.

Hop count is used instead to pick the upstream link the rover watches, to detect route flips (a change with no deployment), and to stop deployment when no relays are left.

Why RSSI is a threshold, not a score term

	RSSI	Score Q
What it tells you	Cause: how strong the radio link is	Outcome: are maps arriving?
Where measured	Rover, in real time	Base; reaches the rover late
Timing	Leading: falls before failure	Lagging: falls after maps are missed
Fit with map delivery	Weak overall (ρ = 0.30)	It is map delivery

- **A weight would blur the warning.** A 10 dB fall would barely move Q, and the lead time would be lost.
- **It would hide real failures.** Scenario 35 had strong RSSI (−63 dBm) but delivered only 45 % of maps.
- **It has no fixed scale.** RSSI differs about 15 dB between Pi boards and is measured opposite to the map flow, and the cliff position is uncertain. A threshold with a margin copes; a normalised score term would carry the error into every score.

Quality score equation

The score gives 75 % of its weight to the map, because the map is what the base cannot replace.

```
Q = 0.50\,S_{gap} + 0.25\,S_{lat} + 0.15\,S_{tfpdr} + 0.10\,S_{tflat}
```

Each term runs from 0 (bad) to 1 (good).

S_gap is the share of time spent at each missed-map level, multiplied by that level's value:

Maps missed in a row (k)	Value	Meaning
0	1.0	Full marks
1	0.7	Target, still acceptable
2	0.3	Inside the hard limit, but weak

Maps missed in a row (k)	Value	Meaning
3 or more	0	Failure

k = 2 gets 0.3 because it is allowed, so it cannot be 0. A window spent wholly at k = 2 must still land below the 0.5 "failing" line.

The other terms are straight lines between two anchors:

Term	Score 1 at	Score 0 at	Why these anchors
S_lat (map latency median)	⩽ 300 ms	⩾ 3,000 ms	3 s is one map period; later than that, the map missed its slot
S_tfpdr (TF PDR)	⩾ 99 %	⩽ 85 %	Proposed TF limit: PDR ⩾ 95 %
S_tflat (TF latency median)	⩽ 100 ms	⩾ 1,000 ms	Proposed TF limit: latency ⩽ 200 ms

All one-hop scenarios that met the map target also meet the proposed TF limits.

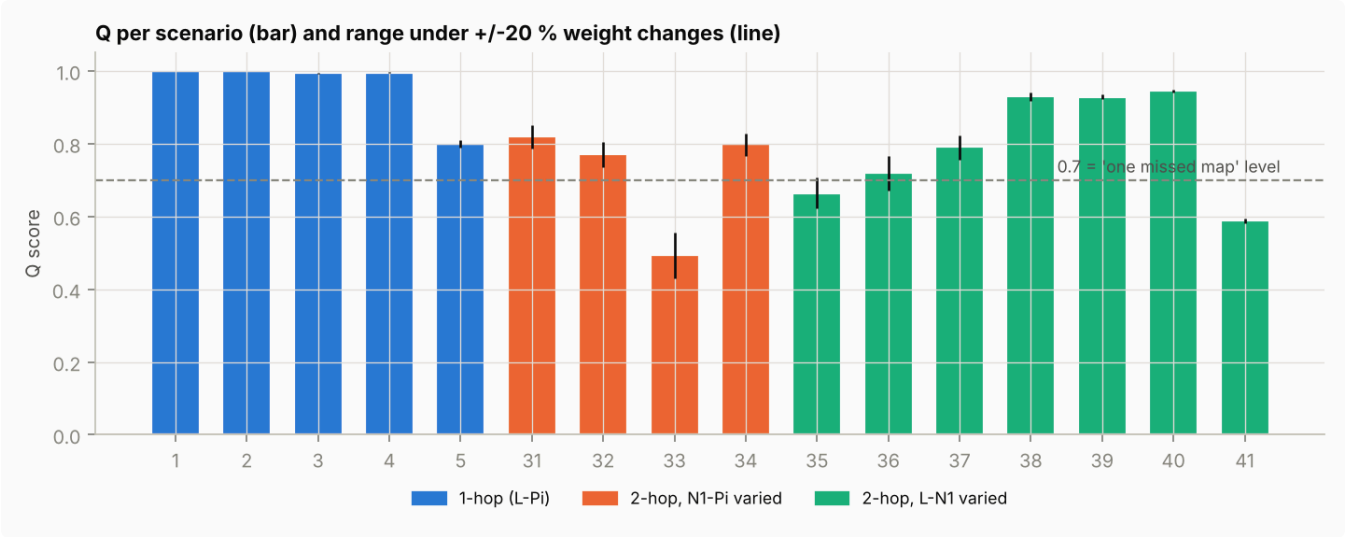
Reading Q: Q ⩾ 0.7 is at or better than the "one missed map" level; Q < 0.5 is failing. Always report the maximum k next to Q, because Q is an average and can hide a short failure. Scenarios 5, 31 and 34 broke the hard limit but still scored 0.80.

Weights compared with objective methods

Parameter	Ours	AHP	CRITIC	Entropy	MEREC
Map gap	0.50	0.48	0.43	0.51	0.18
Map latency	0.25	0.27	0.15	0.05	0.43
TF PDR	0.15	0.16	0.23	0.24	0.19
TF latency	0.10	0.09	0.19	0.19	0.20

AHP, CRITIC and Entropy all rank the map gap first. MEREC ranks map latency first, because it rewards whichever value varies most, and one-hop map latency spans 206–702 ms. AHP's consistency ratio is 0.005 (below 0.1 is acceptable). Objective methods measure how much a value varies, not how much it matters, so they serve as a cross-check rather than the answer.

Scores and sensitivity



Changing every weight by up to ±20 % (2,000 random trials) barely changed the ranking: Kendall $\tau \geq 0.97$, and no scenario moved more than 2 places. Scores from AHP, CRITIC and Entropy weights stay within 0.07 of ours. MEREC scores the two-hop scenarios up to 0.24 higher, because two-hop map latency stays low even when maps are lost.

The ranking makes physical sense: one hop up to 28 m scores 1.0, scenarios 38–40 about 0.93, scenario 5 and the two-hop failures 0.5–0.8, and scenario 33 (N1 past its cliff) lowest at 0.49.

Deployment rule

The rule uses two signals: an RSSI trigger on the rover for early warning, and a map-age check from the base as a backstop. Neither signal alone was enough in the tests.

Single thresholds tested

A "breach run" is a run with any gap of two or more missed maps.

Signal (10 s mean)	Case 1 (one hop)	Case 3 (two hops, laptop–N1)
RSSI ⩽ −68 dBm	Caught both breach runs; 1 false trigger (the scenario 5 run that kept the target)	Missed 3 of 16 breach runs; scenario 35 failed at −63 dBm

Signal (10 s mean)	Case 1 (one hop)	Case 3 (two hops, laptop–N1)
Combined PDR ⩽ 98 %	Caught both; 1 false trigger	Caught 15 of 16, but base-side only and late
TQ ⩽ 210	Caught 1 of 2	Caught 10 of 16

On one hop, any RSSI threshold from −62 to −70 dBm gives the same result, because no position was tested in that band. On two hops, no single threshold separates good from bad.

The recommended rule

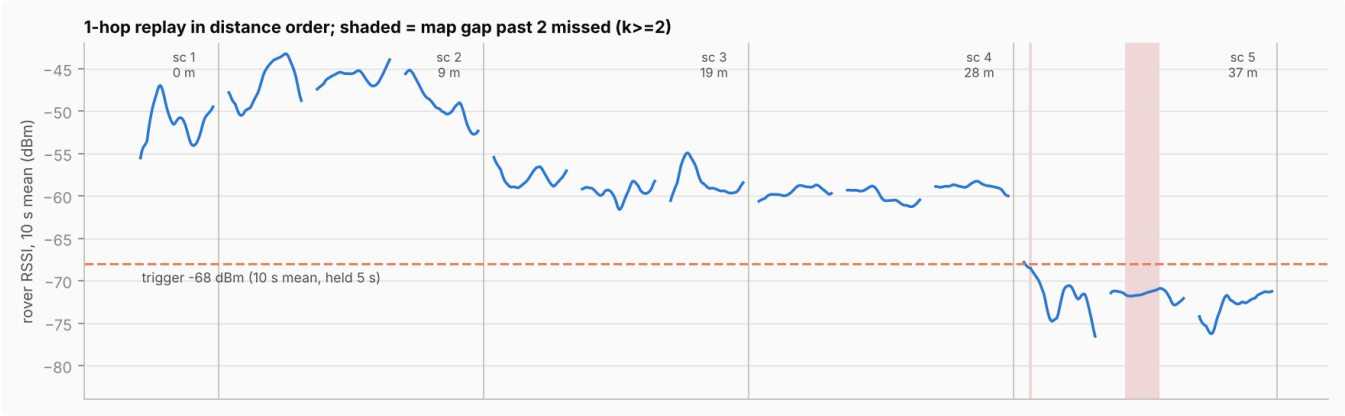
1. **Rover trigger (primary).** If RSSI to the upstream node, averaged over 10 s, stays at or below **−68 dBm for 5 s**, stop and drop the relay here.
   - **Why −68 dBm:** it keeps about 4 dB of margin above the first breach seen (−71.8 dBm). Dropping at −68 to −70 dBm means the relay's own link back starts above the cliff.
   - **Hysteresis:** after a drop, watch the new upstream node. To re-arm after a hold or retreat, the 10 s mean must climb back above −64 dBm.
2. **Base backstop (secondary).** The base sends map age (seconds since the last map) in its heartbeat. If map age is **7 s or more**, one map is missed and the next is late: stop advancing at once. Deploy if no relay is out yet; otherwise retreat until map age clears. A heartbeat silent for 5 s counts as map age ⩾ 7 s.
3. **Base warning (optional, one hop).** If the median map latency over 30 s exceeds 500 ms, slow the rover down.

Decision table

Rover RSSI (10 s)	Base Q / map age	Action
Above −68 dBm	Q ⩾ 0.7 and age < 7 s	Keep exploring
⩽ −68 dBm	Q ⩾ 0.7	**Deploy now.** The intended case: warning before any map is lost
Above −68 dBm	Q < 0.7 or age ⩾ 7 s	**Stop, don't deploy.** The weak link is elsewhere (as in 35–37); retreat until age clears

Rover RSSI (10 s)	Base Q / map age	Action
⩽ −68 dBm	Q < 0.7 or age ⩾ 7 s	Deploy if a relay is left, else retreat
Any	Heartbeat silent ⩾ 5 s	Treat as map age ⩾ 7 s

Replay on the recorded data



- **One hop:** the trigger stays quiet at 0–28 m (RSSI −47 to −61 dBm) and fires at 37 m, the first position where maps were missed. Runs were stationary, so true lead time can't be measured. In a walk, the rover would cross −68 dBm somewhere between 28 and 37 m.
- **Backstop:** fires about 0.5 s before each gap reaches two missed maps. Too late to prevent that gap, but in time to stop the rover before it reaches three.
- **Two hops:** the RSSI rule would not have protected scenarios 31–37, because maps were lost even at good signal. The backstop fired in every breach run.

Success criterion (deploy before two maps are missed): met on one hop at the tested positions. Not reachable on two hops by any rule until the map transport is fixed, for example with a smaller Fast DDS message size, map compression, or a slower map rate.

Limitations and next measurements

The −68 dBm threshold is a margin choice, not a measured edge; the next walk test should pin it down.

Limitation	Effect	Next measurement
No one-hop data between −61 and −70 dBm	Cliff position unknown within ~10 dB	Slow continuous walk from 28 m to 40 m, logging RSSI and map age together
1–5 runs of 60 s per scenario	Per-scenario numbers rest on 1–5 minutes	More runs at the key positions (28–37 m one hop, 26–29 m N1–Pi)
Pi board changed between days	~15 dB RSSI offset; one hop vs two hops not cleanly comparable	Repeat scenarios 4 and 5 on board B
RSSI measured opposite to the map flow	Rover may misjudge its own sending link	Log TX retries and bitrate with `iw station dump` on the sending node
N1–Pi hop not logged in case 3	Can't tell which hop caused 35–37 losses	Log RSSI on both hops at once
Delivery rose with time in 35–40	Distance and time are mixed	Repeat case 3 in reverse order (40 first)
Fragmentation is a hypothesis	Could shift every threshold	Set Fast DDS max message size to ~1,400 B or compress the map, and repeat one scenario
Stationary runs only	Lead time and flapping untested	Walk test with the rule running live